PARAM SHARMA

AI/ML Engineer

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PROFESSIONAL SUMMARY

Computer Science graduate and AI/ML Engineer specializing in Computer Vision, LLM fine-tuning, and end-to-end MLOps architectures. Authored multiple peer-reviewed publications and open-source datasets, bridging the gap between state-of-the-art research (Swin Transformers, custom optimization) and production deployment. Proficient in PyTorch, TensorFlow, Python, C++, and cloud native MLOps tools, dedicated to shipping scalable high-accuracy AI systems.

WORK EXPERIENCES

AI/ML Engineer | Full-time — May 2025 - Present

MaxBrain Technologies — *On-site • Jaipur, India*

Plant Disease Detection System (FasalSaathi): Link

- Curated and published two open-source agricultural image datasets for Ridge Gourd pathology with **9,000+ downloads** and Brinjal pathology.
- Benchmarked deep learning architectures (VGG19, ResNet, InceptionNet) and engineered a hybrid pipeline using a pre-trained DenseNet121 as a feature extractor paired with an XGBoost classifier, boosting accuracy from **78% to 92%**.
- Deployed the optimized inference model into FasalSaathi, a live Android application on the Google Play Store, and secured an Indian Design Patent for an IoT-based automated agricultural data collection device.

Smart EdTech Platform (MCQGyaan AI): Link

- Engineered an enterprise EdTech platform used by over **5,000 students** featuring an AI-driven evaluation engine capable of semantically grading handwritten and typed answers using OCR and NLP.
- Curated custom educational datasets to fine-tune Large Language Models (LLMs) for context-aware assessment, significantly reducing educator grading time.
- Developed full-stack, role-based portals featuring automated question paper generation and real-time student performance analytics.

Technologies / Skills Used : *Python, PyTorch, CNN, LLM Fine-Tuning, NLP, OCR, REST API, React, MongoDB*

PROJECTS

IMDb Sentiment Analysis: Cloud-Native MLOps Pipeline | Github — 2026

- Engineered an IMDb sentiment analysis pipeline achieving **95% accuracy** on **50K+ reviews**, orchestrating multi-stage DAG workflows via DVC and AWS S3.
- Tracked experiments via MLflow & DagsHub, enforcing **5+ automated CI/CD checks** in GitHub Actions for zero-touch model promotion to production.
- Deployed containerized Flask API across **2 EKS pod replicas**, achieving **<50ms latency** with Prometheus scraping and live Grafana dashboards.

Technologies / Tools Used : *AWS EKS, MLflow, DVC, Prometheus, Grafana, Docker, GitHub Actions, NLP*

Brain Tumour Classification (Swin-GWO-Net++) | Github — 2026

- Developed the Swin-GWO-Net++ hybrid transformer model as a final-year Bachelor's thesis, **achieving 99.54% classification accuracy** across **102,600** BraTS 2021 brain MRI scans.
- Engineered a Dual Attention Fusion Module (DAFM), **resulting in a 2.74% accuracy increase** over standard backpropagation in ablation studies.
- Implemented Grad-CAM++ feature mapping to provide visual explainability and clinical interpretability for deep learning model predictions.

Technologies / Tools Used : *Python, Transformers, Deep Learning, PyTorch, Computer Vision*

SKILLS

Languages :	English, Hindi
Databases :	PostgreSQL, MySQL, MongoDB
Programming Languages :	C++, Java, Python, JavaScript, TypeScript, SQL
Frameworks & Libraries :	Django, Flask, FastAPI, Next.js, React, Tailwind CSS
AI, LLMs and Deep Learning :	PyTorch, TensorFlow, Scikit-learn, OpenCV, RAG, Agentic AI, LLM Engineering & Fine-Tuning, Hugging Face, vLLM, Unsloth, LangGraph
Tools & Platforms :	AWS, Docker, Kubernetes, CI/CD, GitHub Actions, MLflow, Prometheus, Grafana, n8n, Model Deployment, Streamlit, Git, GitHub, Postman

EDUCATIONS

Birla Institute of Technology Mesra | Ranchi — Nov 2022 - Jun 2026

Bachelor of Technology (B.Tech) in Computer Science and Engineering — *CGPA : 8.3/10*

University of Tokyo | Japan — Apr 2026 - Jul 2026

Matsuo Lab GCI Program

Selected for the Global Creator Intensive (GCI) — a 3-month AI/ML and Data Science program at Matsuo Laboratory, University of Tokyo. Covered deep learning theory, large-scale array computation, time-series analysis, unsupervised learning, and participated in global data competitions.

AWARDS & ACHIEVEMENTS

- Authored **2 Peer-Reviewed AI/ML Research Papers**: Published main author research at WCAIAA 2026 (TinyML MobileNet1D quantized to **<1.5 MB** with **92.77% accuracy**) and co-authored plant disease diagnosis research in Springer AIIS (MIND 2025).
- Published Open-Source Datasets: Created and released the Ridge Gourd Pathology Dataset on IEEE DataPort (**9,000+ downloads**, **335+ ratings**, featured in IEEE Aug 2025 Newsletter) and the Brinjal Pathology Dataset on Mendeley Data.
- Active contributor to production AI and ERP ecosystems; submitted merged PRs/bug fixes across @frappe and @miracodeai addressing backend pipelines and agent orchestration.
- **Intellectual Property (IP)**: Granted a Government of India Design Patent for an IoT-based automated agricultural data collection device.